BACKEND / DATA ARCHITECTURE SPECIFICATION

Professional AC Core Cutting Business Website

Backend Schema Document • v1.0 • Production Baseline

ITEM	SPECIFICATION
System	AC Core Cutting Business Website
Architecture	Next.js server-rendered / hybrid application with API or server-action enquiry handling
Primary backend purpose	Secure lead capture + centralized business content + extensible service data
Primary data domains	Business profile, services, service areas, gallery, reviews, FAQs, enquiries, analytics
Security priority	Server-side validation, spam protection, secret isolation, minimal personal data
MVP boundary	No customer accounts, payments, complex scheduling or large CMS

Schema principle: keep the MVP small, normalized and maintainable, while leaving clean extension points for future scheduling, lead management, WhatsApp workflows and CMS functionality.

This specification distinguishes requirements explicitly present in the project documents from implementation-level schema decisions proposed to realize those requirements.

1. Backend Scope & Design Principles

The approved technical architecture places the Next.js application between the browser and the lead destination. Public pages use server rendering/static content where useful, while enquiry submission passes through a server/API boundary with validation and spam protection. ■filecite■turn4file0■L29-L45■

The TRD defines a content model for Business, Service, GalleryItem, Review, FAQ and ServiceArea, plus a lead API containing name, phone, WhatsApp preference, service type, location and message. ■filecite■turn4file6■L237-L252■

Backend objectives

• Centralize business-specific content so public pages do not hard-code changing business data.

• Provide a stable source for services, gallery, reviews, FAQs and service areas.

• Accept quote/enquiry requests securely over HTTPS.

• Track the approved conversion events without collecting unnecessary sensitive information.

• Support future content-management or lead-management capabilities without forcing them into the MVP.

• Keep public reads fast and predictable; keep write operations restricted to server/API boundaries.

Important boundary

A full admin/CMS platform is explicitly outside the MVP unless ongoing content management requires it. The schema therefore supports structured content without requiring an admin UI in version 1.0.

2. Proposed Backend Architecture

The TRD specifies Next.js + TypeScript + Tailwind, Zod or equivalent validation, Next/Image, server/API forms and production cloud hosting with HTTPS/CDN. ■filecite■turn4file0■L18-L28■

Recommended logical architecture

> ***Browser → Next.js UI → Server/API or Server Action → Validation → Spam Protection → Database / Lead Destination → Business Follow-up***

LAYER	RESPONSIBILITY	DATA ACCESS
Public UI	Render pages and collect user actions	Read public content; submit enquiry
Next.js server	Routing, server rendering, API/server actions	Controlled reads/writes
Validation	Validate and sanitize untrusted input	Enquiry payload only
Data layer	Persist structured business/content/lead data	Parameterized queries / ORM
Lead delivery	Email / approved CRM / business workflow	Enquiry payload
Analytics	Measure approved conversion events	Minimal event data
Storage/CDN	Serve optimized gallery/media	Public media only

Database engine note

The provided TRD does not mandate a specific database engine. A relational database such as PostgreSQL is a strong implementation choice for the proposed schema, but this is an implementation recommendation rather than a source-mandated requirement.

3. Data Domains & Entity Inventory

ENTITY	ROLE	MVP?	MAIN RELATIONSHIPS
business_profile	Single source of business identity/contact data	Yes	Referenced by public content/settings
services	Structured service catalogue	Yes	Gallery / leads may reference service
service_areas	Genuine served locations	Yes	Leads may reference selected area
gallery_items	Real project media and metadata	Yes	Optional service/category relation
reviews	Approved customer testimonials/public review references	Yes	Independent content entity
faqs	Customer questions and answers	Yes	Optional service relation
enquiries	Quote/contact leads	Yes	Optional service/service-area relation
analytics_events	Approved conversion tracking	Yes	Optional page/service context
site_settings	Small technical/site-level settings	Optional	Deployment/content configuration

The first seven content entities directly reflect the TRD content model; enquiries reflect the documented lead API; analytics_events operationalizes the documented conversion-event requirements. ■filecite■turn4file6■L237-L252■ ■filecite■turn4file5■L206-L210■

Schema philosophy

• Use UUID primary keys for mutable entities where the chosen database supports them.

• Use stable slugs for public service/service-area URLs.

• Keep publication/approval status explicit instead of deleting content unnecessarily.

• Use created_at / updated_at timestamps consistently.

• Use foreign keys where a relationship is meaningful; keep external URLs as URL/text fields where no local entity is required.

4. ERD / Relationship Model

The following is the proposed relational relationship model derived from the approved content model and lead requirements.

```
BUSINESS_PROFILE
■
■■■ SERVICES ■■■■■■■■■■■■■■■■
■ ■ ■
■ ■■■ GALLERY_ITEMS ■
■ ■■■ FAQS ■
■ ■■■ ENQUIRIES ■
■ ■
■■■ SERVICE_AREAS ■■ ENQUIRIES
■
■■■ GALLERY_ITEMS
■■■ REVIEWS
■■■ FAQS
■■■ ANALYTICS_EVENTS
```

Relationship rules

RELATIONSHIP	CARDINALITY	RULE
Business → Services	1 : N	Business can expose many services.
Business → Service Areas	1 : N	Only genuinely served locations are published.
Service → Gallery Items	1 : N / optional	A gallery item may be tied to a service when known.
Service → FAQs	1 : N / optional	Use for service-specific questions; global FAQs may remain unlinked.
Service → Enquiries	1 : N / optional	Selected service is stored when provided.
Service Area → Enquiries	1 : N / optional	Selected location is stored when it maps to a published area.
Business → Reviews	1 : N	Reviews are independently approved content.
Business → Analytics Events	1 : N	Events support conversion measurement.

The schema intentionally avoids a many-to-many join table for services and gallery/FAQ unless implementation later proves that one item must belong to multiple services. This keeps the MVP simpler.

5. business_profile Table

The TRD Business content model identifies business name, logo, phone, WhatsApp, address, hours, city, service areas and description. ■filecite■turn4file6■L237-L244■

FIELD	TYPE	NULL	CONSTRAINT / PURPOSE
id	uuid	No	PK
business_name	varchar(160)	No	Verified public business name
logo_url	text	Yes	Public logo asset URL/path
phone	varchar(32)	No	Primary public phone
whatsapp	varchar(32)	Yes	WhatsApp number if supplied
address	text	Yes	Only if applicable / supplied
city	varchar(100)	No	Main city
hours	jsonb / text	Yes	Business hours when supplied
description	text	Yes	Approved business description
google_business_url	text	Yes	Public review/profile URL if available
created_at	timestamptz	No	Record creation
updated_at	timestamptz	No	Last modification

Data rule

Business-specific phone, WhatsApp, address, service areas, hours and claims must come from the business owner. The TRD explicitly identifies these as implementation inputs. ■filecite■turn4file3■L144-L150■

Implementation note

For a single-business website, business_profile can contain exactly one active row. A future multi-business architecture could introduce a tenant/business_id model, but that is not required by the current project.

6. services Table

The approved service structure includes AC Core Cutting, RCC Core Cutting, AC Drain Hole, Concrete Wall Drilling, Pipe & Cable Passage and Other Services. ■filecite■turn3file4■L257-L271■

FIELD	TYPE	NULL	CONSTRAINT / PURPOSE
id	uuid	No	PK
slug	varchar(120)	No	UNIQUE; public route segment
name	varchar(160)	No	Display name
summary	varchar(300)	No	Card/overview summary
description	text	No	Full service description
benefits	jsonb	Yes	Structured benefit list
process	jsonb	Yes	Ordered process steps
image_url	text	Yes	Primary service image
seo_title	varchar(180)	Yes	Unique indexable title
seo_description	varchar(320)	Yes	Unique metadata description
is_published	boolean	No	Default true/false per workflow
display_order	integer	No	Ordering
created_at	timestamptz	No	Creation timestamp
updated_at	timestamptz	No	Modification timestamp

Indexes

UNIQUE(slug); INDEX(is_published, display_order). Optional full-text/search indexes can be added later if a search feature is introduced.

Service integrity

• Do not publish a service unless the business actually offers it.

• Keep service slugs stable after launch to protect SEO.

• Do not store public pricing unless standardized pricing rules are confirmed and the business wants pricing published. ■filecite■turn4file3■L144-L150■

7. service_areas Table

The service-area model exists to establish local relevance and prevent misleading coverage. The project requires the main city and nearby locations genuinely served. ■filecite■turn3file4■L257-L271■

FIELD	TYPE	NULL	CONSTRAINT / PURPOSE
id	uuid	No	PK
slug	varchar(120)	No	UNIQUE; URL-safe identifier
name	varchar(120)	No	Area display name
description	text	Yes	Optional local context
map_reference	text	Yes	Optional map identifier/link
is_published	boolean	No	Published coverage flag
display_order	integer	No	Ordering
created_at	timestamptz	No	Creation
updated_at	timestamptz	No	Modification

Rules

• Only genuinely served locations are stored as published service areas.

• Avoid generating thin/fake location pages for SEO.

• A lead may retain the customer's entered location even when it is not a published service area; the system should not silently convert arbitrary user input into a public location entity.

• Use a stable slug if a location page becomes indexable.

8. gallery_items Table

The TRD defines GalleryItem fields as image, altText, title, category, location and optional date. ■filecite■turn4file6■L237-L244■

FIELD	TYPE	NULL	CONSTRAINT / PURPOSE
id	uuid	No	PK
image_url	text	No	Optimized public media URL/path
alt_text	varchar(300)	No	Accessible image description
title	varchar(180)	Yes	Optional display title
category	varchar(80)	No	e.g. AC Core Cutting / RCC / Drain Holes
location	varchar(120)	Yes	Job location when appropriate
service_id	uuid	Yes	FK → services.id
project_date	date	Yes	Optional work date
is_published	boolean	No	Public visibility
display_order	integer	No	Gallery ordering
created_at	timestamptz	No	Creation
updated_at	timestamptz	No	Modification

Media rules

• Use real project photographs wherever possible.

• Optimize images before serving; the TRD requires responsive modern formats and appropriate display sizing. ■filecite■turn4file5■L180-L188■

• Never present stock/AI imagery as actual business work.

• Alt text is required for informative images and should describe the actual image.

9. reviews & faqs Tables

The TRD defines Review fields as customerName, reviewText, optional rating/source/date and approved; FAQ fields are question, answer, category and order. ■filecite■turn4file6■L237-L244■

reviews

FIELD	TYPE	NULL	RULE
id	uuid	No	PK
customer_name	varchar(120)	No	Approved display name
review_text	text	No	Actual testimonial/review
rating	smallint	Yes	1–5 when known
source	varchar(60)	No	e.g. manual / Google
source_url	text	Yes	Public review/profile URL
review_date	date	Yes	When known
approved	boolean	No	Must be true for public display
created_at	timestamptz	No	Creation
updated_at	timestamptz	No	Modification

FAQs

FIELD	TYPE	NULL	RULE
id	uuid	No	PK
question	varchar(300)	No	Customer-facing question
answer	text	No	Concise approved answer
category	varchar(80)	Yes	Optional grouping
service_id	uuid	Yes	FK for service-specific FAQ
display_order	integer	No	Ordering
is_published	boolean	No	Public visibility
created_at	timestamptz	No	Creation
updated_at	timestamptz	No	Modification

Review rule: never invent testimonials. Public display requires approval and appropriate attribution/permission. ■filecite■turn3file9■L452-L471■

10. enquiries Table — Core Lead Schema

The enquiry table is the most important write-side entity. The TRD explicitly specifies POST over HTTPS and suggests name, phone, WhatsApp preference, service type, location and message. ■filecite■turn4file6■L245-L252■

FIELD	TYPE	NULL	CONSTRAINT / PURPOSE
id	uuid	No	PK
name	varchar(120)	No	Required lead name
phone	varchar(32)	No	Required contact number
whatsapp_preference	boolean	Yes	Whether WhatsApp is preferred
service_id	uuid	Yes	FK → services.id
location	varchar(160)	No	Customer-provided service location
message	text	Yes	Requirement/context
status	varchar(30)	No	new / contacted / quoted / closed / spam
source_page	varchar(255)	Yes	Page where form was submitted
created_at	timestamptz	No	Lead creation
updated_at	timestamptz	No	Status/change timestamp

Recommended constraints

• name length: sensible maximum.

• phone length: sensible maximum and server-side format validation.

• location length: bounded to prevent abuse.

• message length: bounded to prevent oversized payloads.

• status constrained to a known set.

• No password, government ID, payment information or other unnecessary sensitive fields.

Indexes

INDEX(created_at DESC); INDEX(status, created_at DESC); INDEX(service_id, created_at DESC). If a lead dashboard is added later, these indexes support common operational views.

11. analytics_events Table

The technical requirements call for page views and key conversion events: phone_click, whatsapp_click, quote_start, quote_submit, map_click and service_cta_click. ■filecite■turn4file5■L206-L210■

FIELD	TYPE	NULL	PURPOSE
id	uuid	No	PK
event_name	varchar(60)	No	Approved event name
page_path	varchar(255)	Yes	Current route
service_id	uuid	Yes	Optional service context
metadata	jsonb	Yes	Non-sensitive event context
occurred_at	timestamptz	No	Event time

Allowed MVP events

page_view, phone_click, whatsapp_click, quote_start, quote_submit, map_click, service_cta_click

Privacy rules

• Do not place phone numbers, message contents or other unnecessary personal information inside analytics metadata.

• Keep metadata minimal and documented.

• Use a controlled event_name allowlist so arbitrary client input cannot create uncontrolled event types.

• The TRD explicitly requires privacy-conscious analytics and avoidance of unnecessary sensitive data. ■filecite■turn3file0■L41-L45■

12. Optional site_settings Table

A small settings table is optional. It is useful for non-content technical switches but should not become a premature CMS.

FIELD	TYPE	PURPOSE
key	varchar(80)	Unique setting key
value	jsonb	Typed configuration value
updated_at	timestamptz	Modification timestamp

Possible examples

• analytics_enabled

• contact_form_enabled

• maintenance_message

• default_social_image

• schema_version

Do not store secrets in site_settings. Secrets belong in the deployment environment/secret manager and must never be shipped to client-side code. The TRD explicitly prohibits exposing secrets/API keys/private configuration in client code. ■filecite■turn4file5■L189-L197■

13. API / Server Contract

The backend write API should follow the TRD's POST-over-HTTPS model with validation, spam protection, safe responses and operational logging. ■filecite■turn4file6■L245-L252■

ENDPOINT	METHOD	AUTH	PURPOSE	RESPONSE
/api/enquiries	POST	Public + abuse controls	Create lead	201 success / 400 validation / 429 rate limit / 500 safe error
/api/analytics/events	POST	Public + allowlist	Record approved events	204/200 accepted / 400 invalid

Public content reads

Public service, gallery, review, FAQ and service-area data should normally be loaded through server-side data access rather than exposing a general-purpose public database API.

Example enquiry payload


```
{"name":"Customer Name","phone":"","whatsappPreference":true,"serviceId":"","location":"Area /
City","message":"AC installation wall opening required"}
```


Response contract

Success: return a generic confirmation and an enquiry reference only if the product needs one. Error: return a safe, user-friendly message without stack traces, SQL errors or internal service details.

14. Validation & Spam Protection

All untrusted form input must be validated and sanitized server-side. The TRD requires rate limiting/spam controls on enquiry endpoints. ■filecite■turn3file0■L24-L32■

CONTROL	IMPLEMENTATION
Schema validation	Zod or equivalent shared server schema
Required fields	name, phone, service type and location according to approved form
Length limits	Apply bounded max lengths to every text field
Phone validation	Normalize/validate accepted format before persistence
Service validation	If service_id is supplied, verify it exists and is publishable/valid
Location handling	Treat user-entered location as untrusted text
Rate limiting	IP/request-based throttling on enquiry endpoint
Honeypot	Optional hidden field for simple bot filtering
CAPTCHA-equivalent	Add only if abuse volume justifies it
Duplicate submission	Prevent repeated form submissions from the same UI action
Safe errors	Generic response; detailed diagnostics stay server-side

Never rely on client-side validation alone. The browser can be bypassed.

15. Security & Privacy Schema Rules

The TRD requires HTTPS, server-side validation/sanitization, secret isolation, rate limiting/spam controls, secure headers and avoidance of unnecessary personal data. ■filecite■turn4file5■L189-L197■

• Production traffic must use HTTPS.

• Database credentials remain server-side and outside client bundles.

• Use parameterized queries or a safe ORM/query builder.

• Do not expose raw database errors to users.

• Store only information required to handle an enquiry.

• Restrict database access to the application/server role.

• Use least privilege for deployment and operational credentials.

• Do not put lead details into analytics event metadata.

• Backups and retention should follow the business's operational/privacy requirements.

• If an admin/CMS is added later, protect it with restricted credentials and separate permissions.

Data classification

DATA	CLASSIFICATION	PUBLIC?	RULE
Business name / phone	Business public	Yes	Verified public business data
Service content	Public	Yes	Published content only
Gallery metadata	Public	Yes	Approved real work only
Reviews	Public	Yes	Approved/attributed
Service areas	Public	Yes	Genuine coverage only
Enquiry name/phone/message	Lead personal data	No	Server-side operational access
Analytics metadata	Operational	Limited	No unnecessary personal data
Secrets/API keys	Confidential	No	Environment/secret manager only

16. Indexes, Constraints & Integrity

TABLE	UNIQUE / INDEX	IMPORTANT CONSTRAINTS
business_profile	Single active record	Required verified phone/city
services	UNIQUE(slug); published/order	Stable public slug
service_areas	UNIQUE(slug); published/order	Only genuine areas published
gallery_items	published/order; service_id	alt_text required; service FK nullable
reviews	approved; source	Only approved rows publicly queryable
faqs	published/order; service_id	Question + answer required
enquiries	created_at; status; service_id	Bounded input; known status values
analytics_events	occurred_at; event_name	Allowlisted event names
site_settings	UNIQUE(key)	No secrets

Foreign-key behavior

• Service deletion should normally be soft/unpublish rather than hard delete once public URLs exist.

• Gallery/service and FAQ/service relationships should use safe deletion behavior so content cannot become corrupted.

• Historical enquiries should retain the selected service reference where possible; if a service is retired, preserve the lead record.

• Analytics records should be append-oriented and not updated as normal content.

17. Seed / Initial Data Model

The database should be seeded only with verified project data. The documents identify the following service/content categories but leave business-specific facts pending.

ENTITY	INITIAL DATA TO SEED	STATUS
Business	Name, logo, phone, WhatsApp, city, address/hours if supplied	Pending owner data
Services	Six approved service categories	Verify actual offerings
Service Areas	Main city + genuine nearby locations	Pending owner data
Gallery	Real work images + metadata	Pending real assets
Reviews	Approved testimonials/public review references	Pending approval
FAQs	Core cutting / hole size / RCC / duration / area questions	Draft + verify

The Project Brief explicitly lists the required content assets, including business name/logo, phone/WhatsApp, city/service area, actual services, experience, real work photos, customer reviews, hours and Google Business Profile/review link. ■filecite■turn3file9■L462-L471■

18. Future Schema Extensions

The current schema should leave room for the roadmap without implementing roadmap complexity in the MVP.

FUTURE FEATURE	POSSIBLE ENTITIES / CHANGES	MVP ACTION
Appointment requests	appointments table + availability rules	Do not implement now
Lead dashboard	users/admin_users + enquiry assignments + notes	Keep enquiry status extensible
WhatsApp workflows	workflow events / message templates	Keep phone/WhatsApp data clean
Quote calculator	pricing_rules / service_options	Do not publish pricing without stable rules
CMS	content revisions / authoring / publishing	Current content tables become CMS basis
Multi-language	localized content tables/JSON	Keep slugs and content fields adaptable
Blog / SEO	posts, categories, SEO metadata	Future module
Review automation	review_requests / workflow logs	Future module

The official roadmap includes appointment/request scheduling, automated WhatsApp workflows, admin lead dashboard, quote calculator, review-request workflow, multilingual support and blog/knowledge content. ■filecite■turn1file2■L605-L614■

19. Backend Project Structure

The TRD specifies an application structure containing app routes, API, components, content, lib, styles, types and public assets. ■filecite■turn3file1■L68-L88■

```
src/
■■■ app/
■ ■■■ api/
■ ■ ■■■ enquiries/
■ ■ ■■■ analytics/events/
■ ■■■ ... public routes
■■■ lib/
■ ■■■ db/
■ ■■■ validation/
■ ■■■ spam/
■ ■■■ lead-delivery/
■■■ types/
■■■ content/
■■■ components/
```

Suggested backend responsibilities

• db: database client, schema/migrations and repository/query functions.

• validation: shared Zod schemas for API boundaries.

• spam: rate-limit and bot-protection utilities.

• lead-delivery: email/CRM/business workflow adapter.

• types: shared TypeScript domain types.

• content: static defaults or content adapters when a database is not used for every public page.

The exact ORM, migration tool and email provider are implementation choices and are not mandated by the provided TRD.

20. Backend Acceptance Criteria

- All public content entities can be represented without hard-coded duplication.
- Services have stable slugs and verified publish state.
- Service areas represent only genuine coverage.
- Gallery items support real image, alt text, category and optional service/location metadata.
- Reviews require approval before public display.
- FAQs support global and service-specific content.
- Enquiries accept only the approved useful lead fields.
- Enquiry input is validated and sanitized server-side.
- Enquiry endpoint uses HTTPS and rate limiting/spam protection.
- Private credentials and API keys never reach client-side code.
- Safe generic error responses are returned to users.
- Operational failures can be logged without storing unnecessary sensitive information.
- Analytics supports the approved conversion events without unnecessary personal data.
- Database indexes support recent leads, lead status and service-based reporting.
- Schema supports future CMS/lead-dashboard expansion without requiring a redesign.
- Production database access follows least privilege.
- All schema migrations are version-controlled and reproducible.
- Seed data contains no fabricated business information.
- Backend is tested for validation, failure states, security and data integrity.

21. Final Backend Schema Lock

The backend should remain intentionally simple for the MVP:

> ***BUSINESS PROFILE + SERVICES + SERVICE AREAS + GALLERY + REVIEWS + FAQS + ENQUIRIES + ANALYTICS***

with the central production flow:

> ***PUBLIC PAGE → SERVER DATA ACCESS → VERIFIED CONTENT***

and for leads:

> ***QUOTE FORM → HTTPS POST → VALIDATE → SPAM PROTECTION → PERSIST / DELIVER → SUCCESS***

The schema is designed to support the project's approved priorities: performance, local SEO, maintainability, accessibility, security and conversion. The TRD explicitly states those as the technical architecture priorities. ■filecite■turn4file3■L144-L151■

Important implementation distinction

This document contains a proposed implementation schema where the source documents define requirements but do not prescribe exact SQL tables, database engine, ORM or migration tool. Those implementation choices should be confirmed before coding.

BACKEND SCHEMA SPECIFICATION v1.0 • Production Baseline